

Robust feature aggregation network for lightweight and effective remote sensing image change detection

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ABSTRACT

In remote sensing (RS) image change detection (CD) task, many existing CD methods focus more on how to improve accuracy, but they usually have more parameters, higher computational costs, and heavier memory usage. Designing lightweight and performance-sustainable CD model that is more compatible with real-world applications is an urgent problem to be solved. Therefore, we propose a lightweight change detection network, called as robust feature aggregation network (RFANet). To improve representative capability of weaker features extracted from lightweight backbone, a feature reinforcement module (FRM) is proposed. FRM allows current level feature to densely interact and fuse with other level features, thus accomplishing the complementarity of fine-grained details and semantic information. Considering massive objects with rich correlations in RS images, we design semantic split-aggregation module (SSAM) to better capture global semantic information of changed objects. Besides, we present a lightweight decoder containing channel interaction module (CIM), which allows multi-level refined difference features to emphasize changed areas and suppress background and pseudo-changes. Extensive experiments carried out on four challenging RS image CD datasets illustrate that RFANet achieves competitive performance with fewer parameters and lower computational costs. The source code is available at <https://github.com/Youzhihui/RFANet>.

1. Introduction

Change detection, as one of the favored tasks in remote sensing image processing, focuses on identifying disparities between bi-temporal or multi-temporal images of the same geographical region, which are taken at different periods, densely segmenting each pixel into changed or unchanged categories. As one of the crucial technologies in remote sensing tasks, CD is widely applied in the domains of natural disaster evaluation (Xu et al., 2019), urban planning (Chen and Shi, 2020a), environmental investigation (Ye et al., 2021) and land monitoring (Lv et al., 2022). Therefore, plenty of CD methods have popped up to tackle real-world problems recently.

Traditional CD methods in the early stage primarily utilize machine learning algorithms to analyze and detect changed content by processing hand-designed coarse features at medium- or low-resolution RS image pairs. Algebra-based CD approaches generate change maps by performing algebraic operations on bi-temporal images, often containing change vector analysis (CVA) (Bovolo and Bruzzone, 2006) or image regression (Sun et al., 2021, 2022). The weakness is that suitable thresholds need to be modulated. Other transform-based approaches employ principal component analysis (PCA) (Wu et al., 2021) to transform the feature space, further improving detection performance by

widening the gap between interested and background information of bi-temporal RS images. But they cannot provide a detailed description of changed types in the areas of interest. The post-classification approaches (Ahlqvist, 2008) can tackle this “from-to” problem. In spite of these endeavors, manually crafted rough features merely tend to contain low-level details but lack high-level semantic information, thereby considerably limiting the performance and generalization of model, as well as being susceptible to natural environmental factors.

Nowadays, considerable works have been devoted to integrating deep learning into the CD scope due to its dependable and efficient capability in extracting features. Based on the different phases of information fusion, these methodologies could be broadly described into two streams: single-stream and dual-stream approaches. For the former, aiming to align with the semantic segmentation paradigm, single-stream approaches (Daudt et al., 2018a; Song et al., 2021; Zhao et al., 2023) wrap bi-temporal image pairs into a unified whole for input to segmentation model, but they encounter a limitation in extracting meaningful deep features information from individual images, hampering their ability to effectively discern changed objects. In contrast, dual-stream methods (Daudt et al., 2018a; Zhang et al., 2020; Chen

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et al., 2022b; Zhao et al., 2023; Li et al., 2023) usually leverage Siamese backbones with shared-weight as feature extractors to individually process bi-temporal RS images. Specifically, some methods (Daudt et al., 2018a; Zhang et al., 2020) employ convolutional neural network (CNN) as encoder to capture local semantic information, and then operate on bi-temporal features by decoder to predict the changed maps. To further expand the receptive field, deeper CNN backbones (Zhang et al., 2023a) and dilated convolution (Li et al., 2023) are exploited to generate more robust changed features. Local information fails to represent the whole content in RS images, so attention mechanism (Chen and Shi, 2020b; Shi et al., 2021) is also applied to focus on global change information for improving the CD performance. With the popularity of transformers, some dual-stream CD approaches (Chen et al., 2022b; Bandara and Patel, 2022; Zhang et al., 2023c) model helpful long-range relationships by using pure transformer to extract hierarchical features. Moreover, additional methods (Liu et al., 2023; Yan and Jiang, 2023) integrate the strengths of both CNN and transformers to further enhance the representative capability of features, enabling the model to effectively and thoroughly detect changed regions with different sizes.

Although outstanding performance is exhibited by the aforementioned methods, they have large model sizes, high computational costs, and heavy memory usage. Therefore, it is great significance to develop lightweight CD networks that are easy to deploy on mobile devices (satellites or unmanned aerial vehicles) for meeting practical applications. Some lightweight CD methods (Lei et al., 2023; Codegoni et al., 2023) simultaneously capture spectral and spatial features of bi-temporal images. For example, Codegoni et al. (2023) directly exploited low-level features with smaller and faster network to obtain coarse difference change maps, but lacked deep high-level semantic information to guide decoding. In addition, Siamese-based features to achieve multi-level feature aggregation are accounted for in a few lightweight works (Li et al., 2023), which can improve the weaker feature representation of changed objects extracted by lightweight backbone. There is no doubt that multi-level feature aggregation has proven to be effective in some studies (Wang et al., 2023; Jiang et al., 2023). We think that missing fine-grained details or semantic information in original features output by poor-performing backbone are hardly recoverable. However, densely multi-level feature aggregation to some extent can alleviate it and further improve the performance. To explore the impact of global semantic information, some approaches perform complex attention mechanisms (Feng et al., 2022) or use spatial pyramid pooling (Liu et al., 2021a). However, they may suffer from computational inefficiency or easily ignore small or tiny changed objects.

In this work, a novel lightweight CD approach, called as robust feature aggregation network (RFANet), is proposed. Firstly, to maintain network lightweight and performance feasibility, mobile backbone MobileNetV2 (Sandler et al., 2018) is used as extractor to obtain original bi-temporal multi-level features. Then, to achieve fine-grained details and semantic information complement each other, lightweight and effective feature reinforcement module (FRM) is designed to enhance representation of features between backbone extraction and feature difference operation. Compared to A2Net (Li et al., 2023), FRM uses dense feature aggregation to more thoroughly complement hierarchical features. Next, semantic split-aggregation module (SSAM) is used on difference features. It splits high-level features to mine global distribution information of all changed objects with various scales from different aspects. In addition, channel interactive module (CIM), occupying only a negligible number of parameters, is introduced to strengthen the representation of difference features, which emphasizes critically changed objects while suppressing interference from the background. Finally, change maps obtained using decoding unit are used to gradually calculate total loss with ground truth in a deep supervision strategy.

The main contributions of this work are as follows:

- A novel lightweight CD method RFANet is proposed, which can achieve better performance with fewer parameters by adopting deep supervision strategy.
- Feature reinforcement module (FRM) is designed to densely aggregates multi-scale features extracted by the backbone for enhancing the representative capability.
- Semantic split-aggregation module (SSAM) is used on difference features to obtain globally distributed objects, and channel interaction module (CIM) is introduced to allow multi-scale feature interaction to emphasize changed objects and suppress background and pseudo-change.
- Experimental results on four challenging CD datasets illustrate that RFANet achieves better performance compared to other methods, further validating its effectiveness.

2. Related work

2.1. Traditional CD methods

Traditional CD methods mostly utilize algebra-based, transform-based, and post-classification machine learning approaches to discover and detect changed areas, heavily relying on hand-designed coarse features of clustering or threshold segmentation. Algebra-based approaches typically employ CVA (Johnson and Kasischke, 1998; Bovolo and Bruzzone, 2006) or image regression (Sun et al., 2021; Zhao et al., 2021; Sun et al., 2022) to generate change maps. For example, based on the representation of CVA in polar coordinates, Bovolo and Bruzzone (2006) presented an unsupervised change detection framework to systematically analyze and identify both changed and unchanged pixels. Sun et al. (2022) involved creating an image regression technique to address the challenge posed by heterogeneous image pairs. Trouble exists in that algebra-based approaches heavily involve adjusting appropriate thresholds to deem whether or not changes occur. Therefore, some CD approaches adopt transformation schemes to handle this. Transformation-based approaches utilize PCA (Deng et al., 2008; Wu et al., 2021) to transform the feature space, aiming to enhance detection performance by increasing the distinction between changed and unchanged information. Furthermore, to provide detailed descriptions (“from-to”) about changed types in the interest area, some post-classification methods (Ahlqvist, 2008; Wu et al., 2017) have also been extensively developed. However, hand-crafted features predominantly encompass low-level texture and edge information, lacking semantic contents and exhibiting weak robustness, resulting in the inability to accurately localize changed regions. As remote sensing technology advances rapidly, the performance shortcomings of these traditional CD methods become increasingly evident.

2.2. Deep-learning CD methods

On the one hand, deep-learning CD methods roughly fall into two groups, distinguished by their handling of bi-temporal information at different fusion phases: single-stream (Peng et al., 2019; De Bem et al., 2020; Peng et al., 2020) and dual-stream (Lv et al., 2022; Zhang et al., 2020; Lei et al., 2021; Wang et al., 2021; Chen et al., 2022b; Feng et al., 2023; Zhang et al., 2023b). Single-stream approaches adopt the strategy of stacking original bi-temporal RS images to match a single input of segmentation models, which directly detects changed areas within the context of a pure semantic segmentation paradigm. Based on the U-Net model, FC-EF (Daudt et al., 2018a) directly adjusted the number of input channels and depth of network to predict pixels belonging to changed or unchanged categories. Song et al. (2021) proposed a powerful network by incorporating deformable convolution, aiming to enhance model good at geometrical translation capability for easily detecting changed objects. However, these single-stream approaches face limitations in individually extracting helpful deep features from single image, hindering their ability to effectively distinguish the boundaries of changed areas. As a response to this issue, more Siamese dual-stream methods (Fang et al., 2022; Shi et al., 2021; Jiang et al., 2022; Li et al., 2023; You et al., 2023) have been proposed in recent studies.

These approaches usually adopt Siamese encoders, structures whose parameters are shared, as extractors to separately process bi-temporal RS image pairs. Chen et al. (2022c) proposed EGDE-Net to achieve precise boundary segmentation and preserve the structural integrity of change areas by modeling long-range contexts and exploiting the differences between bi-temporal features. To distinguish genuine changes from spurious changes and detect multi-scale changed objects, CSViG (You et al., 2023) employed graph convolutional network and attention mechanism from two dimensions to mitigate the restricted utilization of long-range dependencies, with improved detection of real interested regions. In addition, both single- and dual-stream CD frameworks were discussed and analyzed in the recent TransWCD (Zhao et al., 2023). The efficacy of these dual-stream approaches lies in their ability to amalgamate the extracted features effectively, resulting in a more accurate and comprehensive representation of the changes observed in bi-temporal data.

On the other hand, existing CD methods can also be categorized into CNN-based (Zhang et al., 2023a,b), transformer-based (Chen et al., 2022b; Zhang et al., 2023c), and their hybrids (Li et al., 2022; Liu et al., 2023; Yan and Jiang, 2023) based on different network structures. To broaden receptive field and gain complete content, more CNN-based CD approaches employed deeper backbones (Zhang et al., 2023a) or utilized dilated convolution (Li et al., 2023) to build richer change information for improving CD performance. This means that CD models have more parameters. Recently, the capability of transformers (Feng et al., 2022; Zhang et al., 2023c) to easily handle long-range dependencies has received increasing attention in CD task. For example, SwinSUNet (Zhang et al., 2022), a pure transformer model based on encoder-decoder structure with a U-shape, was presented to enhance the extraction of global contexts across space-time dimensions, highlighting the significance of capturing comprehensive contextual details. But computing attention of transformer consumes heavy computational resources. In addition, some existing methods (Li et al., 2022; Yan and Jiang, 2023) exploited the hybrid of CNN and transformer to simultaneously capture local information and model global content for better suiting the CD task. While these methods have demonstrated excellent CD performance in all kinds of complex and variable scenarios, they are usually bulky and heavy.

2.3. CD methods based on lightweight backbones

Since vision application scenarios on mobile have only limited computational resources, designing lightweight networks becomes an urgent demand. Recently, some lightweight backbones (Tan and Le, 2021; Mehta and Rastegari, 2022) with fewer parameters and higher efficiency have been employed to extract features for various downstream tasks, even though their capacity for representative features may be reduced, meaning weaker feature representations are extracted. Li et al. (2023) designed lightweight CD network A2Net based on backbone of MobileNetV2 (Sandler et al., 2018), combining progressive feature aggregation and supervised attention to improve features and effectively identify changed objects. In their work, Lei et al. (2023) introduced USSFCNet to simultaneously capture spectral and spatial features of bi-temporal images while reducing computational costs and memory usage. To sufficiently explore the difference between bi-temporal images, the faster and smaller TinyCD (Codegoni et al., 2023) made full use of the effectiveness of low-level features. These lightweight CD methods (Chen et al., 2022a; Codegoni et al., 2023) also presented nice performance. Therefore, a more lightweight and effective CD method is proposed in this article.

3. The proposed approach

In this section, overall architecture of our proposed approach, robust feature aggregation network (RFANet), is first briefly described, and then each individual module that makes up the network is sketched in detail.

3.1. Approaches overview

The schematic architecture of the proposed lightweight change detection network RFANet is depicted in Fig. 1, which is built upon the popular encoder-decoder framework. The encoder includes a multi-level feature extractor, feature reinforcement module (FRM), and semantic split-aggregation module (SSAM); while the decoder contains channel interaction module (CIM) and decoding unit (DU).

Encoder: We leverage the lightweight backbone of MobileNetV2 (Sandler et al., 2018) as Siamese extractor to individually obtain bi-temporal multi-level hierarchical features. For samples trained in the RS image CD task, a pair of images are denoted as I_A and I_B , respectively. They are captured in the same region at different periods A and B . The corresponding ground truth with pixel-level annotations is indicated by Y . The bi-temporal hierarchical features of backbone output can be expressed as $\mathcal{F}_m = \{F_m^1, F_m^2, F_m^3, F_m^4, F_m^5\}$, $m \in A, B$.

To reinforce the representative capability of bi-temporal features, FRM is proposed by densely aggregating shallow details and deep semantic information within the changed areas, which can accomplish complementary contents. The multi-level features are allowed to interact and fuse at current resolution, although this process may produce redundant information. Our view is that redundant content can be suppressed, while missing details and semantic information in the original features cannot be recovered. We use the following formula to express it:

$$R_m^i = \text{FRM}_i(\mathcal{F}_m), \quad m \in \{A, B\}, i \in \{1, 2, 3, 4, 5\}, \quad (1)$$

where R_m^i indicates i th level reinforced feature of image acquired at time m . To reduce the computational costs, we adopt a simplified way to obtain difference features, and the formula can be expressed as follows:

$$D^i = \text{S\&A}(R_A^i, R_B^i), \quad i \in \{1, 2, 3, 4, 5\}, \quad (2)$$

where S&A($\cdot$) denotes the process of element-wise subtraction followed by absolute value operation.

Remote sensing image with wide coverage usually contains a large number of associated objects. Complex scenes can be effectively handled by expanding the receptive field to obtain global semantic information and localize changed objects. Therefore, we use semantic split-aggregation module (SSAM) on difference features D^4 and D^5 to obtain global distribution information of objects to fully exploit the changed areas:

$$D_g = \text{SSAM}(D^4, D^5), \quad (3)$$

where D_g denotes the global semantic feature that is used to guide later information interaction and decoding.

Decoder: We introduce a lightweight decoder to implement top-down multi-decoding of features for obtaining change maps. The decoder mainly consists of channel interaction module (CIM) and decoding unit (DU). To detect real changes and suppress pseudo-changes, CIM fuses neighboring difference features in an optimal way. This is achieved by allowing current feature to compute dynamical attention weights with higher-level feature and global semantic feature. Thereby, model assigns varying degrees of attention to different spatial objects, meaning that important changed areas are emphasized and the background is suppressed, the formula is expressed as:

$$D'_i = \text{CIM}(D_i, D_h, D_g). \quad (4)$$

In each decoding unit (DU), refined low-level feature, high-level feature, and global semantic information are aggregated together by element-wise addition and the depthwise separable convolution (DSCov). The combination of global high-level and local low-level features helpfully localize and detect complete changes. Finally, the output prediction and ground truth are used to compute the loss in deep supervision strategy.

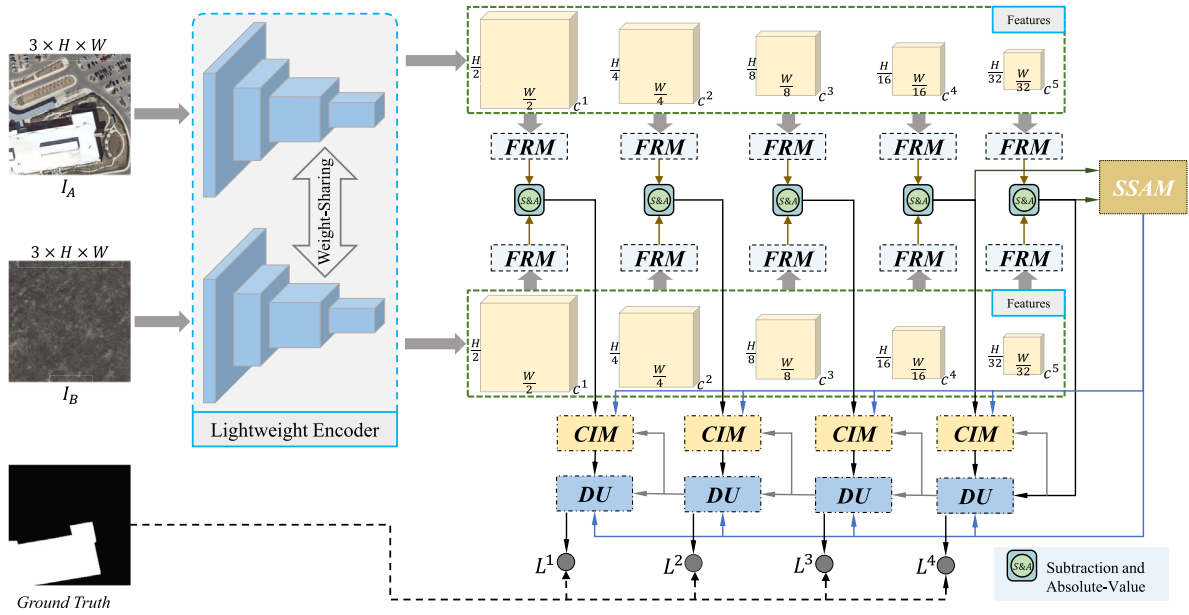


Fig. 1. Overview of the proposed RFANet. Firstly, lightweight weight-shared backbone extracts multi-level bi-temporal features from a pair of images, respectively. Then, FRM is used to allow interactions and fusions between multi-level temporal features for reinforcing representative capability. We use element-wise subtraction and absolute value operations to obtain difference features, and adopt SSAM to better capture global semantic information of changed objects. Next, CIM is used to emphasize changed areas and suppress interference of background and pseudo-changes. Finally, change maps obtained by DU are used to calculate the loss with ground truth in a deep supervision strategy. (H, W) denotes the height and width of original input, and c^i ($i \in \{1, 2, 3, 4, 5\}$) is the number of channels.

3.2. Feature reinforcement module (FRM)

The general consensus is that low-level feature excel in adeptly capturing detailed boundary and texture information of objects, while high-level feature are valued for their ability to hold semantic information crucial for object localization. Given that extracted hierarchical features describe various aspects of changed objects, we propose feature reinforcement module (FRM) that densely integrates multi-level weaker original features extracted by backbone to reinforce the representative capability.

As presented in Fig. 2, the input of each FRM comprises a set of multi-level original features F_m extracted from backbone, where one main feature and four auxiliary features. Here, we exemplify the third FRM applied to reinforce feature from pre-change image I_A , with F_A^3 in set $\mathcal{F}_A$ as the main feature and the others as auxiliary features. Firstly, F_A^3 undergoes a 3×3 convolutional layer with a batch normalization and a ReLU activation function to adjust its channels. Features at lower-level than F_A^3 (here refers to F_A^1 and F_A^2) are downsampled to same resolution as F_A^3 by a max-pooling operation, followed by a 1×1 convolutional layer to adjust their number of channels, and then a 3×3 DSConv to capture local content and filter out noise. Similarly, features at higher-level than F_A^3 (here refers to F_A^4 and F_A^5) first pass through a 1×1 convolutional layer to align number of channels, followed by a 3×3 DSConv layer with a batch normalization and a ReLU activation function, and then a bilinear upsample operation is performed to resize resolution for matching the main features F_A^3 . The entire process of feature conversion can be expressed as:

$$\begin{aligned} R_A^{(3,1)} &= \text{DSConv}^{3 \times 3}(\text{Conv}^{1 \times 1}(\text{MP}(F_A^1))), \\ R_A^{(3,2)} &= \text{DSConv}^{3 \times 3}(\text{Conv}^{1 \times 1}(\text{MP}(F_A^2))), \\ R_A^{(3,3)} &= \text{Conv}^{3 \times 3}(F_A^3), \\ R_A^{(3,4)} &= \text{Up}(\text{DSConv}^{3 \times 3}(\text{Conv}^{1 \times 1}(F_A^4))), \\ R_A^{(3,5)} &= \text{Up}(\text{DSConv}^{3 \times 3}(\text{Conv}^{1 \times 1}(F_A^5))), \end{aligned} \quad (5)$$

where $\text{MP}(\cdot)$ represents max pooling operation, the batch normalization and ReLU activation function are omitted. $R_A^{(3,j)}$ indicate features that have same resolution as F_A^3 from the original j th feature by Eq. (5).

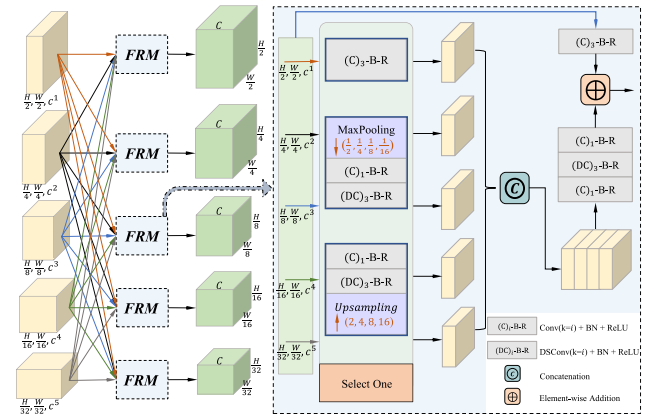


Fig. 2. Overview of feature reinforcement module (FRM). Dense fusion between multi-level features reinforces representative capability.

Following feature conversion, a concatenation operation is employed to fuse features, effectively integrating multi-level information. Subsequently, residual connection is used to maintain the content of F_A^3 , ensuring that $R_A^{(3,j)}$ acts as valuable complement, which can be expressed using the following equation:

$$R_A^3 = \text{Conv}^{3 \times 3}(F_A^3) + \text{Conv}^{1 \times 1}(\text{DSConv}^{3 \times 3}(\text{Conv}^{1 \times 1}(\text{Cat}[R_A^{(3,1)}, \dots, R_A^{(3,5)}])), \quad (6)$$

where $\text{Cat}[\cdot]$ indicates concatenation operation, batch normalization and ReLU activation function are omitted. Each FRM adopts similar operations as above, and the only difference is that main feature and auxiliary features are different. As a result, we gain multi-level bi-temporal reinforced features $\mathcal{R}_m = \{R_m^1, R_m^2, R_m^3, R_m^4, R_m^5\}$, $m \in A, B$. To reduce the computational costs, we directly let the bi-temporal reinforced features at same level take the absolute value after element-wise subtraction, and multi-level difference features are obtained by using Eq. (2), it can be denoted $\mathcal{D} = \{D^1, D^2, D^3, D^4, D^5\}$.

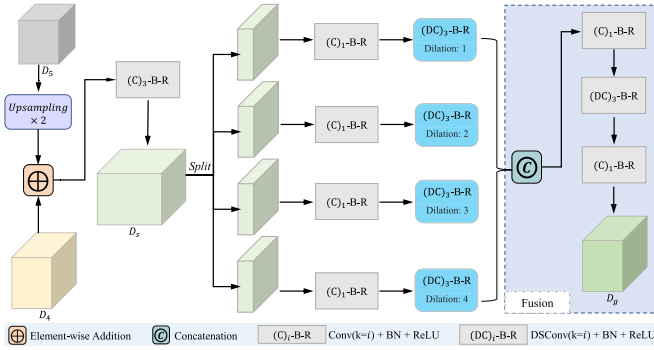


Fig. 3. Overview of semantic split-aggregation module (SSAM). Global semantic changes are aggregated by atrous convolutions with diverse atrous rates.

3.3. Semantic split-aggregation module (SSAM)

High-level features commonly include semantic information that is essential for object localization. To better capture the global semantic changes, the semantic split-aggregation module (SSAM) is constructed as shown in Fig. 3, which uses a series of atrous convolutions to perform multi-scale feature learning. Remote sensing images with wide coverage usually contain more massive objects than traditional nature images, so the highest two-level difference features are aggregated to capture complete and accurate global semantic changes. Specifically, D^5 performs a bilinear upsampling operation to resize resolution as D^4 , and then directly element-wise sum with D^4 to generate coarse global semantic feature. Next, a 3×3 convolutional layer is performed to expand the number of channels, and then global semantic feature is split equally into four sub-features in the channel dimension. This procedure can be expressed using the following equation:

$$D_s = \text{Split}(\text{Conv}^{3 \times 3}(D^4 + \text{Up}(D^5))), \quad (7)$$

where $D_s = [D_s^1, D_s^2, D_s^3, D_s^4]$, batch normalization and ReLU activation function also are omitted, and $\text{Split}(\cdot)$ represents the reverse of concatenation operation in channel dimension.

To model global semantic changes from various scales, these sub-features pass through dilated convolutions with diverse atrous rates to expand the receptive field, and learn multi-scale features as shown in Fig. 3, the formula is expressed as follows:

$$D_{s,g}^i = \text{DSConv}^{3 \times 3}(\text{Conv}^{1 \times 1}(D_s^i)), \quad i \in \{1, 2, 3, 4\}. \quad (8)$$

After that, it is essential to fuse the multi-scale global information to obtain a more complete and refined global content. To keep the model lightweight, 1×1 convolutional layers and 3×3 DSConv layer are used to operate on the concatenated sub-features with different receptive fields,

$$D_g = \text{Conv}^{1 \times 1}(\text{DSConv}^{3 \times 3}(\text{Conv}^{1 \times 1}(\text{Cat}[D_{s,g}^1, \dots, D_{s,g}^4])), \quad (9)$$

where $\text{Cat}[\cdot]$ denotes concatenation operation, batch normalization and ReLU activation function are omitted. In summary, SSAM obtains refined global information with only a few parameters, which can be used to guide decoding operations.

3.4. Decoder module

Considering that low-level features may be susceptible to massive background noise while still containing vital details of changed objects, channel interactive module (CIM) is introduced to enhance the representation of difference features. The wealth of semantic information in high-level and global features exceeds that presented in low-level features, providing an advantage in suppressing background interference. CIM utilizes correlations among features across the different levels

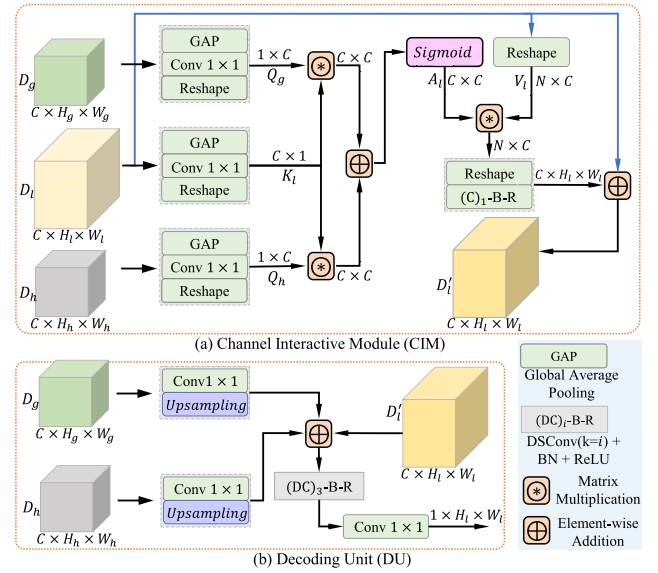


Fig. 4. Overview of decoder module. The low-level ($C \times H_l \times W_l$), high-level ($C \times H_h \times W_h$), and global ($C \times H_g \times W_g$) difference feature are fused in channel interactive module (CIM). Decoding unit (DU) is used to predict change maps.

in channel dimension to emphasize the details of changed areas and improve the representative capability of low-level features.

As shown in Fig. 4(a), the input to CIM consists of three components: low-level feature D_l with size of $C \times H_l \times W_l$, high-level feature D_h with size of $C \times H_h \times W_h$, and global feature D_g with size of $C \times H_g \times W_g$, respectively, and they have same number of channels. Firstly, global average pooling (GAP) and 1×1 convolution operation are performed to project D_l , D_h , and D_g as key K_l , queries Q_h and Q_g . The formula is as follows:

$$\begin{aligned} K_l &= \text{Conv}^{1 \times 1}(\text{GAP}(D_l)), \\ Q_h &= \text{Conv}^{1 \times 1}(\text{GAP}(D_h)), \\ Q_g &= \text{Conv}^{1 \times 1}(\text{GAP}(D_g)), \end{aligned} \quad (10)$$

where $\text{GAP}(\cdot)$ denotes global average pooling operation, the size of K_l , Q_h and Q_g are all $C \times 1 \times 1$. Then, K_l is reshaped to size of $C \times 1$, Q_h and Q_g are reshaped size of $1 \times C$. And K_l performs matrix multiplication operations with Q_h , Q_g respectively to obtain two similarity matrices. Next, the two matrices are added to generate attention weight A_l via using a sigmoid function, which can be expressed using the following equation:

$$A_l = \text{Sigmoid}((\text{Re}(K_l) * \text{Re}(Q_h)) + (\text{Re}(K_l) * \text{Re}(Q_g))), \quad (11)$$

where $\text{Re}(\cdot)$ is reshape operation, $*$ means matrix multiplication. Finally, D_l is reshaped to size of $N \times C$ ($N = H_l \cdot W_l$) and matrix multiplied with A_l , the output is reshaped to size of $C \times H_l \times W_l$ and projected again by 1×1 convolutional layer with a batch normalization and a ReLU activation function. Meanwhile, a residual connection is applied between the output and original low-level difference feature D_l , the formula is as follows:

$$D'_l = \text{Conv}^{1 \times 1}(\text{Re}(\text{Re}(D_l) * A_l)) + D_l. \quad (12)$$

As for the decoding unit (DU) at each level is illustrated in Fig. 1 and Fig. 4(b), its input likewise consists of three elements: refined low-level feature D'_l through CIM, high-level feature D_h , and global features D_g . The high-level and global features undergo 1×1 convolutional layer separately, and then the bilinear upsampling operation is implemented to resize the resolution as low-level feature. The element-wise addition of the three elements is fed into a 3×3 DSConv layer with batch normalization and ReLU activation function to perform fusion and filter

Table 1

Overview information about the four public CD datasets.

Datasets	Image pairs	Image size	Space resolution	Train/Val/Test	Image type	Changed objects
LEVIR-CD (Chen and Shi, 2020a)	637	1024 × 1024	0.5 m/pixel	7120/1024/2048	Satellite images	building
WHU-CD (Ji et al., 2019)	1	32207 × 15354	0.2 m/pixel	5947/743/744	Aerial images	building
CDD-CD (Lebedev et al., 2018)	16 000	256 × 256	0.03–1 m/pixel	10000/3000/3000	Satellite images	building, forest, road ...
SYSU-CD (Shi et al., 2021)	20 000	256 × 256	0.5 m/pixel	12000/4000/4000	Aerial Image	building, vegetation, road ...

out interfering information. Finally, a 1×1 convolution is used to output the prediction result of current level.

$$Y_p = \text{Conv}^{1 \times 1}(\text{DSCConv}^{3 \times 3}(D'_l + \text{Up}(\text{Conv}^{1 \times 1}(D_h)) + \text{Up}(\text{Conv}^{1 \times 1}(D_g)))). \quad (13)$$

3.5. Loss calculation

The CD is commonly treated as binary classification task at the pixel level, and network parameters optimization involves minimizing binary cross-entropy loss. To mitigate the impact of serious training sample imbalance in CD task that has an influence on the performance, dice loss is also optimized. Furthermore, we implement a deep top-down supervision strategy to train the model at each level for more precise and complete prediction map. Mathematically, the loss function of binary cross-entropy can be expressed as:

$$L_{bce} = -\frac{1}{W \cdot H} \sum_{w=1}^W \sum_{h=1}^H (Y^{w,h} \cdot \log Y_p^{w,h} + (1 - Y^{w,h}) \cdot \log(1 - Y_p^{w,h})), \quad (14)$$

where W and H mean the width and height of input RS images, $Y^{w,h}$ and $Y_p^{w,h}$ denote label and prediction of pixel in coordinates (w, h) . It should be noted that $Y_p^{w,h}$ is ensured to be the same size as ground truth $Y^{w,h}$ by a bi-linear interpolation operation. The dice loss can be defined as:

$$L_{dice} = 1 - \frac{2 \cdot |Y \cap Y_p|}{|Y| + |Y_p|}, \quad (15)$$

where $\cap$ denotes the intersection of sets and $|\cdot|$ indicates the size of set. Employing the deep supervision strategy (Zhang et al., 2020; Li et al., 2023), the total optimization loss becomes expressible as:

$$L_{total} = \sum_{i=1}^4 (L_{bce}^i + L_{dice}^i), \quad (16)$$

where i indicates the i th level. During the testing phase, the largest size Y_p , corresponding to half-width and half-height of the original input size, is applied to obtain the final CD result, by using a bi-linear interpolation operation.

4. Experiments

In this section, first four public datasets used in our experiments are profiled. Next, comparative methods and evaluation metrics are presented. Then, experimental environment is stated and hyper-parameters are set. Finally, quantitative results and qualitative visualizations are discussed and analyzed to prove the effectiveness of our network.

4.1. Datasets

Four public change detection datasets are used to assess the effectiveness of RFANet, which are LEVIR-CD (Chen and Shi, 2020a), WHU-CD (Ji et al., 2019), CDD-CD (Lebedev et al., 2018), and SYSU-CD (Shi et al., 2021). A comparative analysis of these datasets is shown in Table 1. Note that all image pairs are high-resolution and captured in the visible light RGB mode.

4.1.1. LEVIR-CD

LEVIR-CD is a public dataset about changed buildings and consists of 637 bi-temporal RS images, whose size is 1024×1024 pixels. These RS image pairs have a spatial resolution of 0.5 m/pixel, and they cover changed building types mainly including apartments, warehouses, residences, and small garages. To facilitate comparing against other CD approaches, all original image pairs according to the given division (7:1:2) are cropped into non-overlapping patches with a size of 256×256 . Finally, we gain 7120/1024/2048 pairs of images for training/validation/test, respectively.

4.1.2. WHU-CD

WHU-CD focuses mainly on changed buildings reconstructed in the same area over several years following an earthquake happened. It comprises a pair of aerial images with a size of 32507×15354 pixels and a spatial resolution of 0.2 m/pixel, and they are photographed in 2012 and 2016. To align with other CD methods, we follow suit by cropping this pair of images with a wide range of scales, obtaining 7434 small image patches with a size of 256×256 . During the experiment, we split all data into 5947/743/744 pairs of images for training, validation, and testing, respectively.

4.1.3. CDD-CD

CDD-CD stands as a challenging CD dataset sourced from Google Earth. It collects 11 high-resolution season-varying bi-temporal satellite image pairs, spatial resolution is 0.03 to 1 m/pixel, with man-made and natural changed areas of various sizes, such as buildings, trees, cars, roads, and forest areas. Similarly keep the data size consistent with other datasets, these image pairs are also cropped into small patches with a size of 256×256 . In total, the dataset possesses 16000 pairs of images, which are divided into 10000/3000/3000 samples for training/validation/test, respectively. We train our model according to the default official division.

4.1.4. SYSU-CD

SYSU-CD includes 20000 pairs of 0.5 m/pixel aerial images captured in Hong Kong from 2007 to 2014. The primary types of changes in this dataset involve new buildings, suburban expansion, vegetation changes, road extensions, etc. Consistent with the other CD datasets, the individual sample size is 256×256 , and we divided all original RS bi-temporal image pairs according to the official ratio of 6:2:2. Finally, we gain 12000/4000/4000 pairs of bi-temporal RS images for training/validation/test, respectively.

4.2. Comparative methods and evaluation metrics

In our experiments, the proposed novel method and other twelve state-of-the-art CD methods are compared, among which there are CNN-based approaches: FC-EF (Daudt et al., 2018b), FC-Di (Daudt et al., 2018b), FC-Conc (Daudt et al., 2018b), DTCDSN (Liu et al., 2021b), SNU-Net (Fang et al., 2022), RDP-Net (Chen et al., 2022a), TinyCD (Codegoni et al., 2023), and A2Net (Li et al., 2023); there are transform-based approaches: ChangeFormer (Bandara and Patel, 2022) and BIT (Chen et al., 2022b); other hybrid approaches: IFNet (Zhang et al., 2020), ICIFNet (Feng et al., 2022), DMINet (Feng et al., 2023), CSViG (You et al., 2023), USSFCNet (Lei et al., 2023), CLAFA (Wang et al., 2023).

Table 2

The quantitative results of different CD methods on LEVIR-CD, WHU-CD, and CDD-CD. The last column indicates the mean value of same report on different datasets. The symbol [†] denotes combination consisting of encoder from USSFCNet (Lei et al., 2023) and our decoder, and [‡] indicates that the backbone of RFANet is pre-trained. Color convention: **best** (red-bold), 2nd-best (blue-underline), and *3rd-best* (orange-italic).

Methods	LEVIR-CD (%)					WHU-CD (%)					CDD-CD (%)					Mean (%)				
	Pre [†]	Rec [†]	F1 [†]	IoU [†]	OA [†]	Pre [†]	Rec [†]	F1 [†]	IoU [†]	OA [†]	Pre [†]	Rec [†]	F1 [†]	IoU [†]	OA [†]	Pre [†]	Rec [†]	F1 [†]	IoU [†]	OA [†]
FC-EF	81.38	66.48	73.18	57.70	97.52	71.63	67.25	69.37	53.11	97.61	79.09	40.73	53.77	36.77	89.90	77.37	58.15	65.44	49.19	95.01
FC-Di	85.43	75.85	80.35	67.16	98.11	47.33	77.66	58.81	41.66	95.63	87.82	48.61	62.58	45.54	91.62	73.53	67.37	67.25	51.45	95.12
FC-Conc	83.85	82.82	83.33	71.42	98.31	60.88	73.58	66.63	49.95	97.04	84.27	47.93	61.10	43.99	91.20	76.33	68.11	70.35	55.12	95.52
DTCDCSN	91.12	87.79	89.43	80.87	98.94	92.57	85.34	88.81	79.87	99.15	92.89	90.06	91.45	84.25	97.57	92.19	87.73	89.9	81.66	98.55
IFNet	91.64	86.72	89.11	80.36	98.90	87.92	81.33	84.50	73.16	98.82	92.03	92.55	92.29	85.68	97.77	90.53	86.87	88.63	79.73	98.5
SNUNet	91.41	88.22	89.79	81.47	98.98	77.05	65.23	70.64	54.61	97.85	91.02	84.17	87.46	77.71	96.52	86.49	79.21	82.63	71.26	97.78
BIT	92.33	87.19	89.69	81.30	98.97	85.07	90.70	87.79	78.24	98.99	89.03	76.87	82.50	70.22	95.30	88.81	84.92	86.66	76.59	97.75
ChangeFormer	91.12	88.64	89.86	81.59	98.98	89.08	80.08	84.34	72.92	98.81	93.39	90.89	92.13	85.40	97.76	91.2	86.54	88.78	79.97	98.52
ICIFNet	91.32	88.64	89.96	81.75	98.99	92.98	85.56	88.32	79.24	98.96	93.41	91.24	92.31	85.73	98.38	92.57	88.48	90.2	82.24	98.78
RDP-Net	89.48	86.94	88.19	78.88	98.81	93.32	85.59	89.29	80.65	99.19	92.78	93.65	93.21	87.29	98.55	91.86	88.73	90.23	82.27	98.85
DMINet	92.52	89.95	90.75	82.99	99.07	93.84	86.25	88.69	79.68	98.97	93.03	92.95	92.99	86.91	98.51	93.13	89.72	90.81	83.19	98.85
CSViG	92.35	90.71	91.52	84.37	99.14	94.69	86.79	90.57	82.76	99.28	94.48	92.33	93.39	87.60	98.11	93.84	89.94	91.83	84.91	98.84
TinyCD	92.83	88.99	90.87	83.26	99.09	91.51	91.79	91.65	84.59	99.34	92.78	90.09	91.42	84.19	97.56	92.37	90.29	91.31	84.01	98.66
USSFCNet	87.67	91.56	89.57	81.11	98.96	89.10	93.08	91.05	83.56	99.30	95.13	95.99	95.56	91.50	98.91	90.63	93.54	92.06	85.39	99.06
CLAFa	91.85	91.78	91.82	84.87	99.17	95.38	92.75	94.05	88.76	99.53	92.78	93.97	93.37	87.56	98.58	93.34	92.83	93.08	87.06	99.09
A2Net	92.08	90.79	91.43	84.21	99.13	96.68	91.78	94.17	88.98	99.55	93.65	92.92	93.29	87.42	98.07	94.14	91.83	92.96	86.87	98.92
Ours	92.32	90.68	91.49	84.32	99.14	95.46	91.68	93.54	87.86	99.50	93.84	92.32	93.07	87.04	98.02	93.87	91.56	92.70	86.40	98.89
Ours [†]	88.80	91.14	89.96	81.75	98.99	88.36	93.41	90.81	83.17	99.29	95.17	96.29	95.72	91.80	98.95	90.78	93.61	92.16	85.57	99.08
Ours [‡]	92.55	91.03	91.78	84.82	99.17	96.42	92.98	94.67	89.88	99.58	93.61	93.25	93.43	87.67	98.11	94.19	92.42	93.29	87.46	98.95

In line with other change detection methods (You et al., 2023), five evaluation metrics are reported in our experiments to determine the quality of different CD methods: intersection over union (IoU), F1 score, precision (Pre), recall (Rec), and overall accuracy (OA), where IoU and F1 score are the primary metrics, while the others are auxiliary metrics.

4.3. Implementation details

We implement the proposed approach with PyTorch framework and conduct training/inference using a single NVIDIA Quadro RTX 3090Ti GPU with 24 GB. In our experiments, we adopt lightweight backbone MobileNetV2 (Sandler et al., 2018) as weight-sharing feature extractor, and then the output multi-scale features are entered into our designed module. We optimize parameters of network by applying the Adam optimizer with momentum as 0.9 and weight decay as 0.0001. We set the initial learning rate as 0.001 and batch size as 32. During training, poly scheduling strategy is employed to slowly adjust the learning rate to 0 until trained for 40000 iterations. At the conclusion of each training epoch, we conduct inference to assess CD performance of model on validation set, and then evaluate the test set as comparable quantitative results by employing this model with the highest F1 score from validation results of all epochs. Furthermore, according to whether the backbone uses pre-trained on the ImageNet, we conduct two groups of comparative experiments. To further boost the capacity of proposed model for generalization, some data augmentation techniques also are employed during training phase, which include random re-scale, cropping, and flipping (horizontal and vertical).

4.4. Quantitative results

The quantitative reports of five evaluation metrics on LEVIR-CD, WHU-CD, and CDD-CD are presented in Table 2, where the top three values are marked using red-bold, blue-underline, and orange-italic. As shown in Table 2, our proposed method achieves the most optimal performance for OA score on LEVIR-CD when the backbone employs pre-trained on ImageNet, and the other four evaluation metrics obtain competitive scores. More detailed description, the F1/IoU/Pre/Rec/OA score of RFANet finishes 91.78/84.82/92.55/91.03/99.17%, exceeds recent CSViG (You et al., 2023) by 0.26/0.45/0.20/0.32/0.03%, and outperforms the A2Net (Li et al., 2023) by 0.35/0.61/0.47/0.24/0.04%. Even if the backbone of proposed method fails to utilize pre-trained, all metrics achieve acceptable scores. Although RFANet and CLAFa (Wang et al., 2023) accomplish competitive CD performance, CLAFa suffers from two times the number of parameters and five times the computational costs than our method (as shown in Table 3). When it comes to WHU-CD, experimental data presented in Table 2 indicate even more

Table 3

The quantitative results (%) of different CD methods on SYSU-CD. The symbol [†] denotes combination consisting of encoder from USSFCNet (Lei et al., 2023) and our decoder, and [‡] indicates that the backbone of RFANet is pre-trained. Color convention: **best** (red-bold), 2nd-best (blue-underline), and *3rd-best* (orange-italic).

Methods	Params	FLOPs	Training time	SYSU-CD (%)				
	(M) ↓	(G) ↓	(H) ↓	Pre [†]	Rec [†]	F1 [†]	IoU [†]	OA [†]
FC-EF	1.35	3.58	3.73	78.70	68.10	73.02	57.50	88.13
FC-Di	1.35	4.73	4.98	90.97	48.65	63.39	46.40	86.75
FC-Conc	1.55	5.33	5.78	80.70	71.70	75.93	61.20	89.28
DTCDCSN	31.26	13.21	3.50	83.87	71.95	77.45	63.20	90.12
IFNet	35.73	82.26	18.87	78.49	78.23	78.36	64.42	89.81
SNUNet	12.03	54.83	9.30	78.04	74.81	76.39	61.80	89.09
BIT	3.50	10.63	12.93	80.08	71.20	75.38	60.48	89.03
ChangeFormer	41.03	202.80	32.13	79.70	76.75	78.20	64.20	89.91
ICIFNet	23.84	25.41	138.27	71.35	48.58	57.80	40.65	83.27
RDP-Net	1.69	27.12	8.08	82.39	77.19	79.70	66.25	90.73
DMINet	6.24	14.55	47.01	64.47	72.04	68.04	51.56	84.04
CSViG	37.96	202.73	32.97	84.43	79.34	81.80	69.21	91.68
TinyCD	0.29	1.54	8.68	82.35	76.90	79.53	66.02	90.67
USSFCNet	1.52	4.86	20.23	76.30	80.10	78.15	64.14	89.94
CLAFa	5.06	16.60	15.17	85.15	79.37	82.16	69.72	91.87
A2Net	3.78	3.05	3.05	86.45	79.67	82.93	70.83	92.26
Ours	2.86	3.16	4.45	82.06	81.36	81.71	69.08	91.41
Ours [†]	0.90	2.96	21.15	80.11	75.40	77.69	63.51	89.15
Ours [‡]	2.86	3.16	4.45	84.83	82.31	83.55	71.75	92.36

obvious improvement in the proposed model about CD performance. In the two primary evaluation metrics F1 and IoU score, our method surpasses A2Net by 0.5 percentage and 0.9 percentage. It is more than two percentage higher than other many CD methods in the F1 score and more than three percentage above in IoU score. As for the experimental results reported on CDD-CD, we can observe from Table 2 that the proposed method also achieves competitive CD performance in terms of F1 and IoU metrics, which are 93.43/87.67%, respectively. The fact is that while USSFCNet (Lei et al., 2023) exhibits excellence on CDD-CD, it presents a really dismal record on LEVIR-CD and WHU-CD. Specially, when the encoder in our RFANet is replaced with that from USSFCNet, our method accomplishes the highest accuracy on each evaluation metric for CDD-CD dataset, which demonstrates the effectiveness of our decoder. In addition, we further report the mean value of each metric on the three datasets to better compare the overall effectiveness of these different methods. The mean scores in this table reveal that F1, IoU, and OA receive the top values, with Pre and Rec also finishing among the top three. In summary, the proposed method achieves excellent results on three public CD datasets with lower computational costs and fewer parameters compared to other state-of-the-art approaches.

The three datasets analyzed above are sufficiently researched in the change detection community, and excellent performance has been achieved by many methods. To further validate the effectiveness of the proposed method, a more difficult dataset, SYSU-CD, is reported. Table 3 presents the quantitative results of different CD methods.

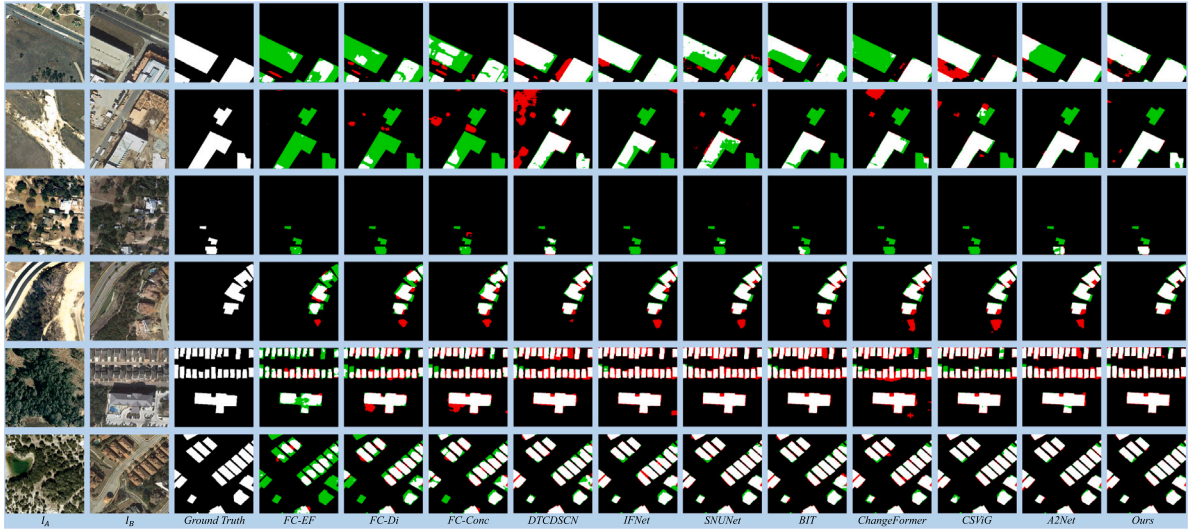


Fig. 5. Qualitative visualizations of different change detection approaches tested on *LEVIR-CD*. The presented colors indicate true positives (white), true negatives (black), false positives (red), and false negatives (green). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

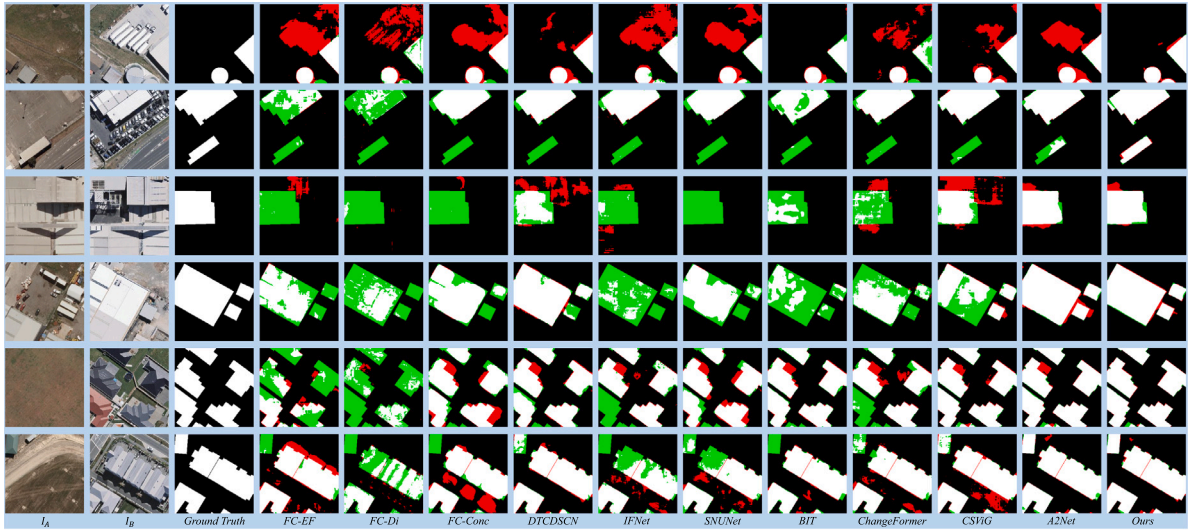


Fig. 6. Qualitative visualizations of different change detection approaches tested on *WHU-CD*. The presented colors indicate true positives (white), true negatives (black), false positives (red), and false negatives (green). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Firstly, we show the parametric quantities (Params), computational costs (FLOPs), and training time of all compared models. As can be seen from the results, parameter number of our model is just 2.86 M, while recent method CSViG has more than ten folds of our parameters, and A2Net likewise has 0.9 M more parameters than our model. As for computational costs, our proposed model reaches the lowest FLOPs compared to other CD methods except for recent A2Net and TinyCD (Codegoni et al., 2023). These data indicate that our model is more lightweight and portable. In addition, the training time also reflects the efficiency of different methods. It can be observed from Table 3 that the training time of our proposed method on the SYSU-CD takes 4.45 h, which is merely higher than the time taken by FC-EF (Daudt et al., 2018b) and A2Net. Next, the reports of five evaluation metrics on SYSU-CD are analyzed. As shown in Table 3, our RFANet achieves the best performance in terms of F1, IoU, Rec, and OA, which are 83.55/71.75/82.31/92.36%, respectively. Moreover, our Pre score is competitive compared to other methods. Although TinyCD and USSFCNet are slightly superior to RFANet in efficiency, the performance of all evaluation metrics is substantially weaker than that of our method.

4.5. Qualitative analysis

To visually assess and compare the detection performance among different CD approaches, we also carry out qualitative visualizations on test sets of LEVIR-CD, WHU-CD, and CDD-CD, which are presented in Figs. 5–7. For better discrimination and observation, different rendering colors are used to indicate true positives (white), true negatives (black), false positives (red), and false negatives (green). The qualitative results presented in these figures illustrate that our method exhibits superiority over other CD methods in the following aspects.

Firstly, the proposed approach is able to precisely locate the real changed objects and detect them. For example, in the third row of Fig. 5, second row of Fig. 6, and first row of Fig. 7, many compared methods fail to identify some small-scale changed objects in the bi-temporal images, which cause the incorrect prediction of false negatives, whereas they are correctly and completely detected by using the proposed method. Secondly, for complex changed scenes, our approach is able to identify them entirely and accurately. Specifically, in the third row of Fig. 6 and fifth row of Fig. 7, although these comparative methods also can locate changed objects, they fail to

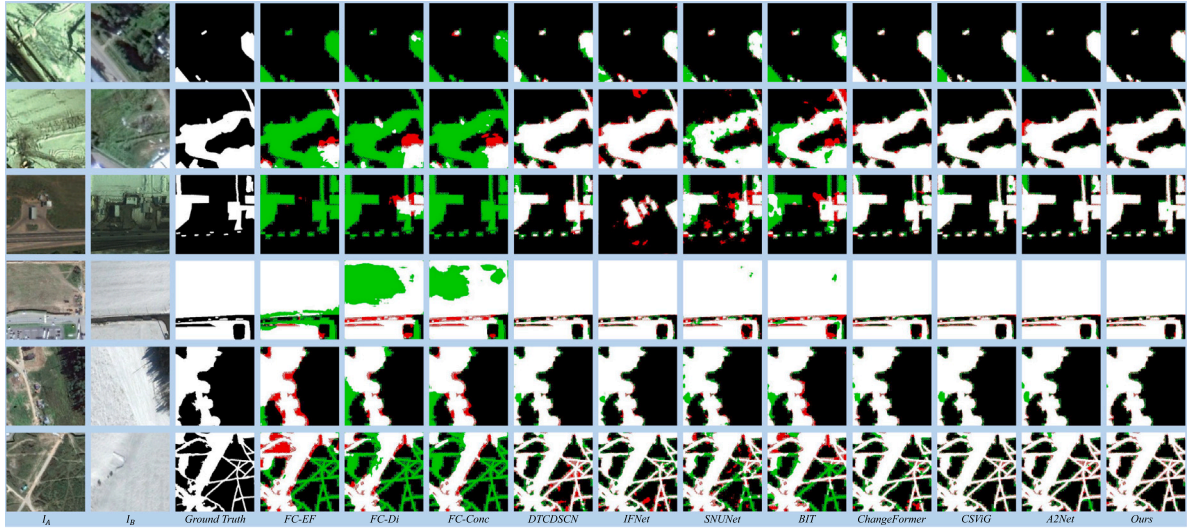


Fig. 7. Qualitative visualizations of different change detection approaches tested on CDD-CD. The presented colors indicate true positives (white), true negatives (black), false positives (red), and false negatives (green). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

detect the entire change areas and boundary information. In the fifth row of Fig. 5, there are dense tiny changed buildings in bi-temporal RS images. Many approaches, such as FC-Di, DTCDSCN, ChangFormer, and A2Net, wrongly predict the background between these dense buildings, while RFANet can reveal the edges of changed buildings to improve its detection performance. In addition, the proposed method is superior in discriminating pseudo-changed objects. In the fourth row of Fig. 5 and first row of Fig. 6, due to the external environmental factors, some areas or large trucks appear similar to the color of buildings, which cause other CD methods mistake such pseudo-changes as real changed objects. Many existing approaches struggle to eliminate these pseudo-changes arising from the diverse appearance of land cover. In contrast, our proposed method is able to deal with them properly and effectively. Apart from changed buildings, CDD-CD provides several other types of change. As presented in Fig. 7, some appearing vegetation in the second row, traveling cars in the fourth row, and substantial disappearing roads in the last two rows, our RFANet is able to identify them accurately and completely, whereas many comparative approaches urgently deserve further improvement.

Therefore, both quantitative experimental data in the previous Section 4.4 and qualitative visualization in this subsection sufficiently illustrate that our proposed CD method exhibits superior performance with lower computational costs and fewer parameters. The main reasons are the use of FRM to densely aggregate multi-level features for enhancing feature capability, as well as the utilization of SSAM to obtain global distribution of changed areas, and the introduction of CIM to emphasize the changed objects and suppress background and pseudo-changes.

5. Discussions

5.1. Ablation analysis

In our proposed CD method, feature reinforcement module (FRM), semantic split-aggregation module (SSAM), and channel interaction module (CIM) are assembled together to improve the accuracy of model while keeping it lightweight. To analyze the contribution of these components, we conduct a comprehensive ablation study on the four datasets. As shown in Tables 4–5, which exhibit the quantitative experimental results about contributions of different components. On the LEVIR-CD dataset, we firstly treat (a) as baseline to train the model without FRM, SSAM, and CIM, the F1/IoU/Pre/Rec/OA score is 90.56/82.76/91.94/89.23/99.05%, respectively. Then, we analyze

the effect of single module by embedding FRM, SSAM, and CIM into (a) to obtain the corresponding results in (b), (c), and (d), respectively. Specifically, the performance is significantly improved when only FRM is used, with all evaluation metrics higher than baseline, where F1 and IoU scores are higher by 0.52% and 0.84%, respectively. When only SSAM is inserted, the F1/IoU/Pre/Rec/OA score is 90.80/83.13/91.72/89.93/99.07%, with some metrics slightly exceeding baseline. As for CIM being used alone, the data in (d) show F1/IoU/Rec/OA score exceeds (a) by 0.37/0.61/1.08/0.03%. The experimental results in (b)–(d) demonstrate that three modules designed by us are effective. Next, we randomly combine two of the modules to perform experiments. We can observe from (e)–(g) in Table 4 that the combination of two modules can further improve CD performance of model than using a single module. Moreover, the accuracy reaches a peak when FRM, SSAM, and CIM are embedded into the overall model. All evaluation metrics F1/IoU/Pre/Rec/OA score achieves 91.49/84.32/92.32/90.68/99.14%, outperforming baseline by 0.93/1.56/0.38/1.45/0.09%. Certainly, similar operations are performed on the CDD-CD dataset. When all components are not employed, the baseline achieves F1/IoU/Pre/Rec/OA score as 91.86/84.94/92.47/91.25/97.68%. With the use of different components individually or in combination, the CD performance improves appreciably. When all components are plugged into the model, F1/IoU/Pre/Rec/OA score reaches 93.07/87.04/93.84/92.32/98.02%, exceeding the baseline by 1.21/2.10/1.37/1.07/0.34%. Quantitative results in Table 4 strongly suggest that each component is useful and effective, our model can accomplish better performance with lower computational costs and fewer parameters. The quantitative experimental results about contributions of different components for WHU-CD and SYSU-CD are presented in Table 5.

Next, different lightweight backbones are used as encoders to further verify the proposed method, such as ResNet18 (He et al., 2016), ShuffleNet (Zhang et al., 2018), EfficientNet (Tan and Le, 2019), and MobileViT (Mehta and Rastegari, 2022). It should be stated that (a)–(h) of Tables 4–5 employ MobileNetV2 as the backbone. As the results presented in Table 4, RFANet with MobileNetV2 beats ResNet18 and EfficientNet in terms of efficiency, but loses to more lightweight ShuffleNet and MobileViT. On the LEVIR-CD dataset, our RFANet with MobileNetV2 achieves the best performance on all evaluation metrics except for Rec. Although using ShuffleNet has lower computational costs and fewer parameters, performance is significantly weaker. Using MobileViT as the backbone has half parameters of using MobileNetV2, but CD performance is slightly worse. As for the CDD-CD dataset, our

Table 4

The quantitative results (%) of our proposed method are reported on *LEVIR-CD* and *CDD-CD* datasets when different variants are set.

Methods	Variants	Efficiency		LEVIR-CD (%)					CDD-CD (%)				
		Params (M) ↓	FLOPs (G) ↓	Pre ↑	Rec ↑	F1 ↑	IoU ↑	OA ↑	Pre ↑	Rec ↑	F1 ↑	IoU ↑	OA ↑
(1) Different components are employed													
(a)	w/o	2.55	1.50	91.94	89.23	90.56	82.76	99.05	92.47	91.25	91.86	84.94	97.68
(b)	FRM	2.70	3.04	92.08	90.10	91.08	83.62	99.10	93.56	91.96	92.76	86.49	97.92
(c)	SSAM	2.64	1.53	91.72	89.93	90.80	83.13	99.07	92.85	91.57	92.21	85.54	97.77
(d)	CIM	2.63	1.61	91.56	90.31	90.93	83.37	99.08	92.87	91.66	92.26	85.59	97.78
(e)	FRM + SSAM	2.79	3.07	92.11	90.34	91.22	83.85	99.11	93.82	92.15	92.98	86.88	97.99
(f)	FRM + CIM	2.79	3.14	92.18	90.41	91.29	83.97	99.12	93.70	91.95	92.89	86.73	97.95
(g)	SSAM + CIM	2.70	1.62	91.79	90.24	91.01	83.50	99.09	92.97	91.71	92.34	85.76	97.80
(h)	FRM + SSAM + CIM	2.86	3.16	92.32	90.68	91.49	84.32	99.14	93.84	92.32	93.07	87.04	98.02
(2) Different backbones are employed													
(i)	ResNet18	12.70	7.91	91.49	91.24	91.37	84.11	99.12	94.21	92.15	93.17	87.21	98.05
(j)	ShuffleNet	0.73	2.50	91.37	89.41	90.38	82.45	99.03	93.38	91.61	92.49	86.02	97.85
(k)	EfficientNet	4.24	3.35	91.51	90.79	91.15	83.73	99.10	93.99	92.07	93.02	86.95	98.01
(l)	MobileViT	1.44	3.01	92.13	90.42	91.27	83.93	99.12	93.90	91.90	92.89	86.73	97.97
(3) Results when FRM is replaced by NAM of A2Net on different backbones													
(m)	MobileNetV2 + FRM	2.86	3.16	92.32	90.68	91.49	84.32	99.14	93.84	92.32	93.07	87.04	98.02
(n)	MobileNetV2 + NAM	2.81	1.38	91.78	90.39	91.08	83.62	99.10	93.83	91.70	92.75	86.48	97.93
(o)	MobileViT + FRM	1.44	3.01	92.13	90.42	91.27	83.93	99.12	93.90	91.90	92.89	86.73	97.97
(p)	MobileViT + NAM	1.34	1.22	91.79	89.55	90.66	82.91	99.06	93.03	91.87	92.45	85.96	97.84

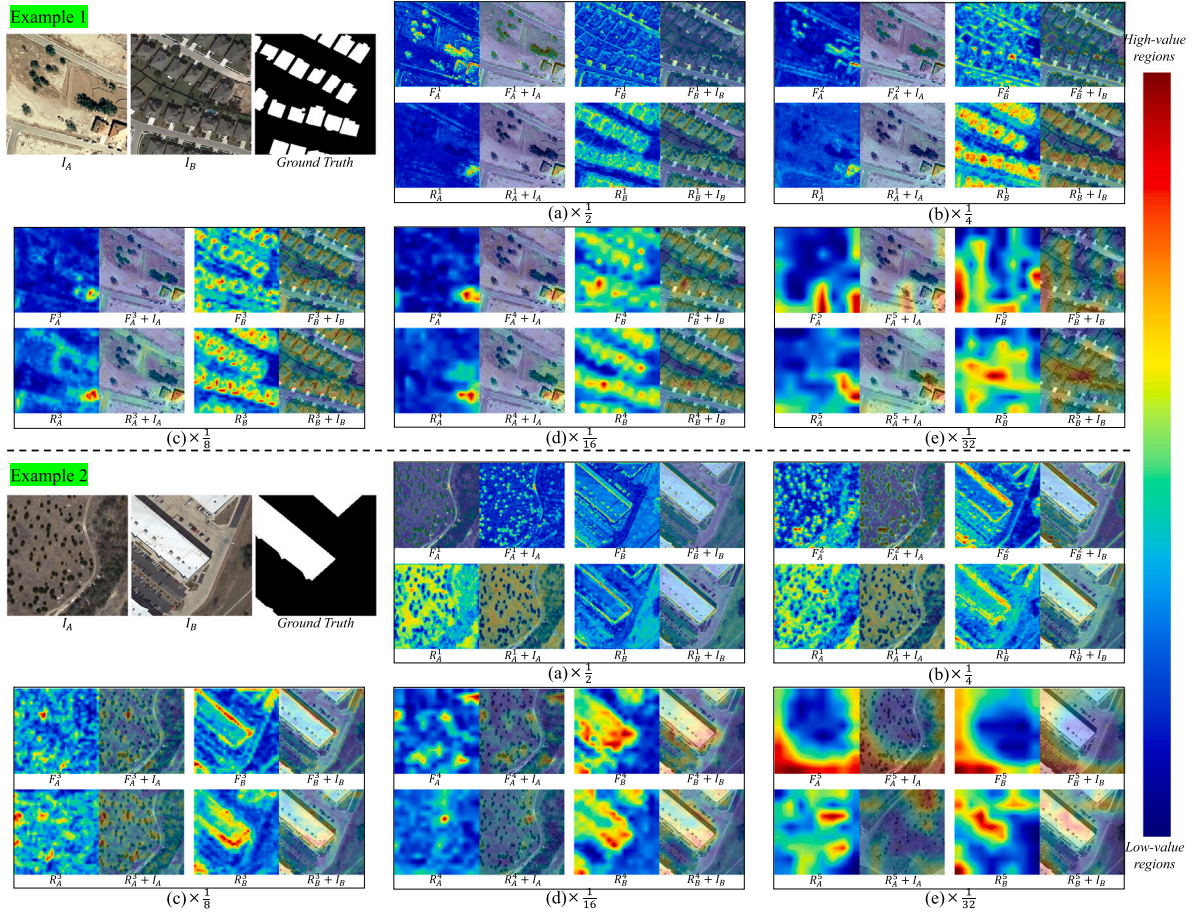


Fig. 8. Heatmap visualizations of multiple levels for RFANet. I_A and I_B are the bi-temporal images followed immediately by corresponding ground truth. (a)–(e) are the results of downsampling $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, and $\frac{1}{32}$, respectively, where F_A^i , $i \in \{1, 2, 3, 4, 5\}$, is original feature extracted by backbone for single image I_A in i th level, $F_A^i + I_A$ denotes the composite of F_A^i and I_A . Similarly, F_B^i and $F_B^i + I_B$ indicate same meaning from image I_B . Comparably, R_A^i is reinforced feature output by FRM for I_A in i th level, $R_A^i + I_A$ denotes the composite of R_A^i and I_A , as well as R_B^i and $R_B^i + I_B$ are analogous implications for I_B .

RFANet with ResNet18 as the backbone manifests an advantage in CD performance. These experimental results demonstrate the effectiveness of lightweight backbones and our components.

Moreover, we further compare FRM with NAM of A2Net to present the performance distinction between dense aggregation and neighbor aggregation. As shown in (n) and (p) from Table 4, the efficiency of model is significantly improved when FRM is replaced by

NAM in our RFANet, but CD performance is degraded. Concretely, when model is trained on LEVIR-CD with MobileNetV2 as backbone, the score of evaluation metric reported by (m) employing FRM is 91.49/84.32/92.32/90.68/99.14%. With NAM instead of FRM, the corresponding experimental data of (n) is 91.08/83.62/91.78/90.39/99.10%, respectively. The F1/IoU/Pre/Rec/OA score accomplished with FRM is higher than that achieved with NAM on LEVIR-CD by 0.41/0.70/0.54/

Table 5

The quantitative results (%) of our proposed method are reported on WHU-CD and SYSU-CD datasets when different variants are set.

Methods	Variants	Efficiency		WHU-CD (%)					SYSU-CD (%)				
		Params (M) ↓	FLOPs (G) ↓	Pre ↑	Rec ↑	F1 ↑	IoU ↑	OA ↑	Pre ↑	Rec ↑	F1 ↑	IoU ↑	OA ↑
(1) Different components are employed													
(a)	w/o	2.55	1.50	94.14	91.64	92.87	86.69	99.44	81.61	78.09	79.81	66.41	90.68
(b)	FRM	2.70	3.04	93.44	92.57	93.00	86.92	99.45	81.30	79.41	80.35	67.15	90.84
(c)	SSAM	2.64	1.53	94.37	91.87	93.11	87.10	99.46	80.21	80.64	80.42	67.26	90.74
(d)	CIM	2.63	1.61	94.24	92.12	93.17	87.21	99.46	82.22	79.16	80.66	67.59	91.05
(e)	FRM + SSAM	2.79	3.07	94.04	92.26	93.14	87.16	99.46	81.52	79.78	80.64	67.56	90.97
(f)	FRM + CIM	2.79	3.14	95.16	91.31	93.20	87.26	99.47	82.64	80.32	81.46	68.72	91.38
(g)	SSAM + CIM	2.70	1.62	94.65	92.15	93.38	87.59	99.48	81.53	81.75	81.64	68.98	91.33
(h)	FRM + SSAM + CIM	2.86	3.16	95.46	91.68	93.54	87.86	99.50	82.06	81.36	81.71	69.08	91.41
(2) Different backbones are employed													
(i)	ResNet18	12.70	7.91	94.00	90.28	92.11	85.37	99.39	79.47	83.03	81.21	68.37	90.94
(j)	ShuffleNet	0.73	2.50	93.65	91.24	92.43	85.93	99.41	78.20	81.53	79.83	66.43	90.28
(k)	EfficientNet	4.24	3.35	93.25	91.83	92.53	86.10	99.41	82.73	79.67	81.17	68.31	91.28
(l)	MobileViT	1.44	3.01	94.01	91.37	92.67	86.35	99.43	80.56	79.74	80.15	66.87	90.68
(3) Results when FRM is replaced by NAM of A2Net on different backbones													
(m)	MobileNetV2 + FRM	2.86	3.16	95.46	91.68	93.54	87.86	99.50	82.06	81.36	81.71	69.08	91.41
(n)	MobileNetV2 + NAM	2.81	1.38	94.29	92.15	93.21	87.28	99.47	81.42	81.41	81.41	68.65	91.23
(o)	MobileViT + FRM	1.44	3.01	94.01	91.37	92.67	86.35	99.43	80.56	79.74	80.15	66.87	90.68
(p)	MobileViT + NAM	1.34	1.22	93.31	91.31	92.30	85.69	99.40	81.27	80.26	80.76	67.73	90.98

0.29/0.04%, while exceeding 0.32/0.56/0.01/0.62/0.09% on CDD-CD. When backbone is switched to MobileViT, (o) and (p) in Table 4 correspond to FRM with dense aggregation and NAM with neighbor aggregation, respectively. The quantitative data from Table 4, both using different backbones and in different datasets, fully justify the superiority of FRM over NAM.

5.2. Visualization and analysis of FRM

To further illustrate the effectiveness of FRM in improving representative capability of features, we employ heatmap visualizations across different levels. As depicted in Fig. 8, two examples are randomly selected from the test set of LEVIR-CD for exhibition. The I_A and I_B refer to bi-temporal images captured at different times A and B in the same region. (a)–(b) denote visualization results of downsampling $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, and $\frac{1}{32}$ features at different levels, respectively. F_m^i indicates original feature extracted by backbone from image I_m at the i th level, where $i \in \{1, 2, 3, 4, 5\}$ and $m \in \{A, B\}$. $F_m^i + I_m$ is the composite of feature F_m^i and image I_m . Following Eq. (1), R_m^i means reinforced feature output by FRM in i th level. As can be seen from first example in Fig. 8, for the low-level features (a) and (b), although the backbone focuses on building information in pre-change image I_A that is really desired to be extracted, it also emphasizes some plants that may affect the detection of changed buildings. Features F_B^1 and F_B^2 can merely concern the local information and edge texture of buildings. However, when FRM is used, a lot of interfering information is suppressed in R_A^1 and R_A^2 , meanwhile the entire building content of post-change image I_B can be completely identified in R_B^1 and R_B^2 . In the second example of Fig. 8 (I_B contains newly constructed buildings), for the high-level features (c), (d) and (e), the features (such as F_B^3 , F_B^4 , F_B^5 in example 2) extracted by backbone for I_B include more interfering noises fail to accurately localize the building information. Nevertheless, once FRM is employed to interact and fuse the multi-level features, it outputs features that precisely localize and contain entire buildings. The qualitative heatmaps in Fig. 8 further demonstrate that FRM is effective and improves the representative capability of model.

5.3. Generalization of methods

To validate the generalization of our proposed method, we compare the CD performance of different approaches when using training and test sets come from different datasets, respectively. The quantitative data are shown in Table 6, which contains two groups of experimental data. In the first group of experiments, models are trained and validated on LEVIR-CD and then tested on WHU-CD afterward. It should be noted

Table 6

The quantitative results (%) of different CD methods about generalization. Color convention: **best** (red-bold), 2nd-best (blue-underline), and *3rd-best* (orange-italic).

Methods	LEVIR-CD _(train) → WHU-CD _(test)					CDD-CD _(train) → WHU-CD _(test)				
	Pre ↑	Rec ↑	F1 ↑	IoU ↑	OA ↑	Pre ↑	Rec ↑	F1 ↑	IoU ↑	OA ↑
FC-EF	18.44	86.05	30.37	17.91	84.35	5.71	6.37	6.02	3.10	92.12
FC-DI	65.31	75.43	70.00	53.85	97.44	14.78	26.42	18.96	10.47	91.04
FC-Conc	51.17	<u>78.22</u>	61.87	44.79	96.18	13.33	17.39	15.09	8.16	92.24
DTCDSCN	51.79	<i>75.95</i>	61.59	44.50	96.24	18.38	63.60	28.52	16.63	87.35
IFNet	61.49	60.24	60.86	43.73	96.93	6.20	25.40	9.97	5.24	81.79
SNUNet	56.72	66.47	61.21	44.10	96.66	5.37	45.74	9.61	5.05	65.86
BIT	61.54	74.33	67.33	50.75	97.14	17.68	<u>78.51</u>	28.86	16.87	84.65
ChangeFormer	23.41	75.05	35.69	21.72	89.27	10.07	52.76	16.91	9.24	79.43
ICIFNet	62.32	74.39	67.82	51.31	97.20	<u>28.02</u>	20.51	23.68	13.43	<u>94.76</u>
RDP-Net	82.59	56.99	67.44	50.88	97.82	25.84	62.21	<u>36.52</u>	<u>22.34</u>	91.43
DMINet	74.60	71.08	<u>72.80</u>	<u>57.23</u>	97.89	28.25	38.76	32.68	19.53	<i>93.66</i>
CSViG	75.07	62.76	68.37	51.94	97.70	15.64	65.23	25.23	14.43	84.66
TinyCD	76.59	60.53	67.62	51.08	97.70	1.22	1.81	3.610	1.81	96.03
USFPCNet	36.26	45.79	40.47	25.37	95.77	22.87	10.58	14.47	7.80	89.27
CLAFa	83.43	64.00	72.44	56.78	98.07	<u>26.93</u>	71.37	39.11	24.31	91.18
A2Net	79.98	66.28	<u>72.49</u>	<u>56.85</u>	<u>98.00</u>	22.48	<u>72.24</u>	34.28	20.69	89.01
Ours	<u>80.38</u>	67.07	73.12	57.63	<u>98.04</u>	23.13	79.96	<u>35.88</u>	<u>21.86</u>	88.66

that the two datasets just mentioned comprise solely changed buildings. The proposed method reaches the best generalization performance on the two primary evaluation metrics, F1 and IoU, which outperforms recent A2Net by 0.63/0.78%, simultaneously beating CLAFa. Other auxiliary metrics also accomplish competitive performance. In another group, the training dataset CDD-CD includes other change types except for buildings. As we can observe from Table 6, the generalization of all methods leaves room for improvement, and we speculate that this is caused by huge discrepancies in the different datasets. However, our method achieves the best result in Rec, and the other evaluation metrics accomplish favorable performance. Overall, the quantitative experimental data reported in this table demonstrate that the generalization of our proposed method outperforms many other CD approaches.

6. Conclusions

Although existing CD methods achieve excellent performance, the models are bulky and clunky. Therefore, we propose a novel lightweight CD method with lower computational costs and fewer parameters in this paper. We first use a lightweight mobile backbone to gain bi-temporal hierarchical features from RS image pairs, respectively. Then, we design a feature reinforcement module (FRM) to densely interact and fuse all multi-level weaker temporal features extracted by mobile backbone for enhancing representative capability. Next, to ensure the lightweight of model, we directly let the bi-temporal features of same level take the absolute value after element-wise subtraction to obtain the difference features, after that we apply semantic split-aggregation

module (SSAM) on high-level difference features to better capture global semantic contents of changed objects. In the decoding phase, we employ channel interaction module (CIM) to enable the model can pay different attention to different spatial objects. Finally, top-down multi-level feature fusion is performed to predict the change maps with fine details and global semantic information. The experimental results conducted on four public CD datasets indicate that our method achieves better CD performance than many previous counterparts.

Although this proposed method is promising, there are still some limitations. Our approach is unable to identify ultra-small changed objects and detect integrity complex changed areas. In addition, the proposed method is sensitive to the boundary of changed regions. In the future, we will design more appropriate approaches for small or ultra-small changed areas, and improve the detection boundary of large and complex changed objects.

CRediT authorship contribution statement

Zhi-Hui You: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Data curation. **Si-Bao Chen:** Software, Resources, Project administration. **Jia-Xin Wang:** Validation, Supervision. **Bin Luo:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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